United Nations Development Programme

Country: Sudan

Initiation Plan

Project Title: Solar for Health: Sustainable Energy Solutions for Equitable Access to Health Services.

UNDP Strategic Plan Primary Outcome:

Relevant Strategic Plan Outcomes and Indicators:

Outcome 1: Growth and development are inclusive, and sustainable, incorporating productive capacities that create employment and livelihoods for the poor and excluded

IRRF Output 1.5: Inclusive and sustainable solutions adapted to achieve increased energy efficiency and/or sustainable energy solutions targeting undiscovered communities/groups and women.

Indicator 1.5.1: Number of new development partnerships with funding for improved energy efficiency and/or sustainable energy solutions targeting underserved communities/groups and women.

Indicator 1.5.2: Extent of change in: a) energy efficiency and/or b) modern energy coverage by users and specific sectors

UNDAF Outcome(s):

UNDAF Outcome 2: Populations vulnerable to environmental risks and climate change become more resilient, and relevant institutions are more effective in the sustainable management of natural resources.

Indicator 1.1: Comprehensive integrated natural resource management framework, including climate change, disaster risk, water, forest and biodiversity management and environmental protection, developed, approved and adopted

Target: At least 5 initiatives supported to develop legal and policy framework related to climate change mitigation/adaptation, water and forest resource management, environmental protection

Expected CP Outcome(s):

Expected CPAP Output (2.2): Investment in green energy and access by needy communities to sustainable energy improved

Executing Entity/Implementing Partner:

- Ministry of Health (MOH)
- Ministry of Water Resources, Irrigation and Electricity (MWRE)
- Higher Council for Environment and Natural Resources (HCENR)

Brief Description

Oil plays a major role in the Sudanese economy and represents 95% of export revenues for the country, while biomass fuels constitute 70% of energy sources and crude oil 30% (Please include evidence) Hydroelectricity is Sudan's largest source of power, accounting for 68% of generation in 2011, followed by diesel and heavy fuel oil (27%) and biomass and waste (5%). Although power generation has more than tripled since the year 2000, millions of people are still without access to electricity. According to the Ministry of Water Resources, Irrigation and Electricity, only about 35% of Sudan's population had access to electricity in 2011. The Sudan Renewable Energy Master Plan was drafted in 2005 in an effort to promote the use of renewable energy sources, including priority projects such as PV (what PV? please spells out in full for first time) installations and biomass co-generation, so as to avoid technological dependence on an oil-based market in energy sector development.

Communities in Sudan are exposed to significantly increased risk of malaria and other major diseases due to effect of climate change (NAPA 2007 and INDC 2015). According to the studies, the results have shown that the risk of transmission potential could increase substantially by 2060. Meanwhile, quality of health services is weakened by the limited access to reliable electrical power across the country remote area. The Sudan's proposed GCF project aims to build resilience to climate change risks in health quality services through a highly innovative set of activities by using access to reliable electricity from solar for subsequent scale-up to other vulnerable communities. The key objective of the proposed project is to increase high quality health services by ensuring the reliable running of temperature and hygrometry controls, adequate lighting systems, refrigeration, cold rooms and IT networks for efficient stock and management information systems in health facilities. These measures affect to minimize and reverse the impact of weather and climate impacts on communities from increasing climatic variability and climate change. In general terms, these initiatives involve a hierarchy of activities that span risk minimization, risk absorption, and risk avoidance strategies in health sector and improve health services to help save the lives of the local people.

Programme Period: April – December -2017 CPAP Programme Component: Atlas Award ID: 00102733 Management Arrangement: DIM PAC Meeting Date: TBD	Total resources required US$150,000 Total allocated resources: US$150,000

Agreed by UNDP:

Name: Selva Ramachandran Date:12/April/ 2017

Title: Country Director

I. PURPOSE

Unreliable electricity access leads to vaccine spoilage, interruptions in the use of essential medical and diagnostic devices, and lack of even the most basic lighting and communications for maternal delivery and emergency procedures. Further, as disease patterns change, even more energy is required to expand services for prevention and treatment of non-communicable diseases. Investments by Governments and donors in health systems, infrastructure and equipment are subsequently unrealized due to the inefficient use of energy technologies.

In Sudan, the biomass fuels constitute 70% of energy sources and crude oil 30% (include source of data as foot note, please). Hydroelectricity is Sudan's largest source of power, accounting for 68% of generation in 2015, followed by diesel and heavy fuel oil (27%) and waste (5%)[1]. Although power generation has more than tripled since the year 2000, millions of people are still without access to electricity. According to the Ministry of Water Resources, Irrigation and Electricity, only about 35% of Sudan's population had access to electricity.

The Sudan Renewable Energy Master Plan was drafted in 2005 in an effort to promote the use of renewable energy sources, including priority projects such as solar systems installation and biomass co-generation, so as to avoid technological dependence on an oil-based market in energy sector development.

Modern health services require stable and reliable energy supply. This is even more important for the supply chain of health commodities where correct temperature conditions are essential for the quality of medicines, diagnostics, equipment and reagents.

II. EXPECTED OUTPUT

The primary objective of the proposed IP is to assess current situation of access to energy in targeted health facilities and formulate the Solar for Health project concept design which include the following outputs:

Output one: A **concept note for GCF Project formulated**

Output two: **A technical Pre-Feasibility Report Prepared**

III. PROJECT PREPARATION ACTIVITIES:

UNDP Sudan will lead the project development process and manage the IP budget. The IP Atlas budget is presented in the Section D "Total Budget and Work Plan". UNDP Sudan will be responsible for selection and recruitment of IP consultants (local and international), developing TORs, arranging IP relevant travel and meetings. Quality assurance and technical advice for the full project development will be provided by the UNDP Regional Hub in Istanbul.

The project preparation includes undertaking the following activities:

1. **Preparation and Submission of a concept note for GCF Project:**

A detailed concept note will be based on a review of available literature, baseline information, consultations with communities and authorities at the targeted states. The project concept note will be prepared as per the "template of the Green Climate Fund". The project concept note should target vulnerable communities who are effected by the climate change across the Sudan.

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1 Source: MINISTRY OF PETROLEUM-Sudanese Petroleum Corporation: Sudan Energy Hand Book 2014.

2. **Preparation and Submission of a draft Technical Pre-Feasibility Report:**
The preparation of a technical pre-feasibility report as per UNDP guideline and GCF best practice, including a thorough situational analysis identifying gaps, needs and barrier profiles and hence the recommended response to these through the project in development. International and national consultants will be recruited to lead the preparation of the concept note.

3. **Arranging a field mission for the technical advisors from UNDP Regional Hubs and the international consultant:**
UNDP will arrange a field mission to Sudan for the technical advisors and the international consultant. The mission will essential capturing the comments on the design of the GCF concept note and the Technical Pre-Feasibility report. Additionally, the International Consultant should identify key stakeholders involved in the project including the private sector, civil society organizations, local and indigenous communities, and their respective roles.

4. **Finalization of the draft GCF concept note and the Technical Pre-Feasibility report:** Incorporation of comments from UNDP, Implementing Partners and UNDP regional Office and Submission of the final concept note and technical pre-feasibility report.

IV. MANAGEMENT ARRANGEMENTS

The project will be directly implemented by UNDP in close partnership with the Ministry of Health with hands-on experience on the technical elements from different stakeholders. The Ministry of Health in coordination with the ministry of Water resources, Irrigation and Electricity will be responsible for the coordination between UNDP and all stakeholders as well as providing the required data on health facilities and energy requirements.

V. MONITORING

A detailed schedule of project reviews meetings will be developed by the project management unit, in consultation with project implementation partners and stakeholder representatives.

Through the project assurance function UNDP in coordination with the implementing partners will monitor and evaluate a set of output and outcome indicators agreed upon through timely data from both routine and non-routine sources and continuous monitoring visits to the project areas. Standard UNDP procedures for monitoring and evaluating projects will be applied.

VI. ANNUAL WORK PLAN

Year: 2017

EXPECTED OUTPUTS *And baseline, indicators including annual targets*	PLANNED ACTIVITIES *List activity results and associated actions*	TIMEFRAME				PLANNED BUDGET	
		Q1	Q2	Q3	Q4	Budget Description	Amount (USD)
Output 1: A concept note for Solar for Health GCF Project formulated	1 Recruitment of consultants		X			71200 - International Consultants	25,000
	2 Preparation of the GCF Prefeasibility study		X	X	X	71300 - Local Consultants	20,000
	3 Travel		X			71600 - Travel	5,000
Output 2: Full Fledge Project Document Developed	4 Recruitment of consultants		X			71200 - International Consultants	30,000
	5 Preparation of the GCF Concept Note		X	X	X	71300 - Local Consultants	40,000
	6 Arranging filed mission to Sudan for RTs		X			71600 - Travel	20,000
	7 Conduct some consultative workshops			X		75700 - Training, Workshops and Conference	10,000
TOTAL							**150,000**